

OM CHAUHAN

 Pune, Maharashtra, 411018  +91 7862909368  omchauhan5505@gmail.com  linkedin.com/in/om-chauhan-7a2b0a290
 github.com/themysterious

PROFESSIONAL SUMMARY

Result-oriented Data Science and Artificial Intelligence student with hands-on experience building, optimizing, and deploying machine learning workflows and production-grade system infrastructure. Proactive builder skilled in Python, cloud deployments, and SRE monitoring. Experienced in automating infrastructure pipelines, configuring secure networking, and developing self-healing operational scripts to optimize application reliability.

EDUCATION

Dr. D. Y. Patil Institute of Technology - *B.E. in Artificial Intelligence and Data Science* Expected June 2027
Pimpri-Chinchwad, Pune, India

SKILLS

Programming & Libraries: Python, JavaScript/TypeScript, SQL, Git, React, Node.js, Express, FastAPI, Flask	DevOps & Cloud (SRE): Docker, Docker Compose, Ansible, AWS EC2, UFW/Firewalls, Nginx, IPFS
AI, NLP & Databases: Llama 3.2, LangChain, ChromaDB, Transformers, BERT, Embeddings, Vector Databases	Monitoring & Analytics: Prometheus, Grafana, Node Exporter, Power BI, Tableau

PROJECTS & EXPERIENCE

DevGuard: Automated Infrastructure Deployment & Intelligent Self-Healing Platform

GitHub (<https://github.com/themysterious/devguard>) Live Demo (<http://13.201.93.49>)

- Deployed a multi-container monitoring and application stack to AWS EC2 using multi-stage Docker builds, minimizing runtime image footprints by 60%.
- Wrote configuration playbooks in Ansible to provision dependencies, set up Nginx reverse proxy, and enforce security policies using host firewall rules (UFW).
- Designed an automated Prometheus and Grafana pipeline to ingest hardware metrics and application stats, providing real-time infrastructure visualization dashboards.
- Built a custom Python Z-score anomaly detector script to analyze metric streams and auto-trigger recovery scripts to restore services during traffic surges.

LexAgent: Multi-Agent Legal Intelligence & RAG System

GitHub (<https://github.com/Rishikesh073/JUDICARY-SYSTEM>)

- Developed a decoupled 3-tier legal intelligence platform featuring an interactive React workspace, an Express orchestration bridge, and a high-performance FastAPI RAG engine.
- Engineered a 4-stage Multi-Agent Orchestration Engine utilizing LangChain and local Llama 3.2 to parse natural language queries, perform semantic searches across 37,000+ Supreme Court judgments, synthesize holding arguments, and run cross-referenced validation.
- Optimized retrieval latency to sub-2 seconds by chunking legal PDFs using Sentence Transformers (all-MiniLM-L6-v2) and persist-indexing embeddings in a local vector vault with ChromaDB.
- Built a real-time event streaming pipeline using Server-Sent Events (SSE) to feed agent telemetry and "Chain of Thought" status updates from the Python engine through Node.js to the frontend visualizer.
- Integrated a Web3 decentralized document vault using IPFS (via Pinata API) to store case memorandums, securing tamper-proof metadata and providing cryptographic Content Identifiers (CIDs) to users.

Smart Energy Trading Controller (ML-Enhanced IoT)

- Engineered an ML-based regression model predicting household energy consumption patterns to automate electrical energy trading decisions.
- Integrated real-time predictive insights with an IoT controller prototype, enabling automated routing of power resources based on expected demand.

QuoraTech Dashboard: AI-Powered Tech News Curation

GitHub (<https://github.com/themysterious/quora->)

- Developed an automated RSS-based content curation pipeline with a Flask backend and React frontend to fetch, summarize, and manage tech news.
- Integrated the Groq API (LLaMA-3.3-70b) to dynamically draft engaging Quora posts and identify relevant user questions from raw RSS feeds.
- Built a real-time Server-Sent Events (SSE) logging system to monitor background AI summarization tasks directly within a Vite/Tailwind CSS dashboard.

ACTIVITIES & ACHIEVEMENTS

- **Leadership:** Treasurer, Data Science & Artificial Intelligence (DSAI) Club — Managed budgeting and organized college tech workshops.
- **Hackathons:** Competed in Vortexa 2.0, Arcadia, Smart India Hackathon (SIH), and Solana Cypherpunk.
- **Certifications:** Python (Data Science), Artificial Intelligence Foundational Track.